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Assignment-2



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1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

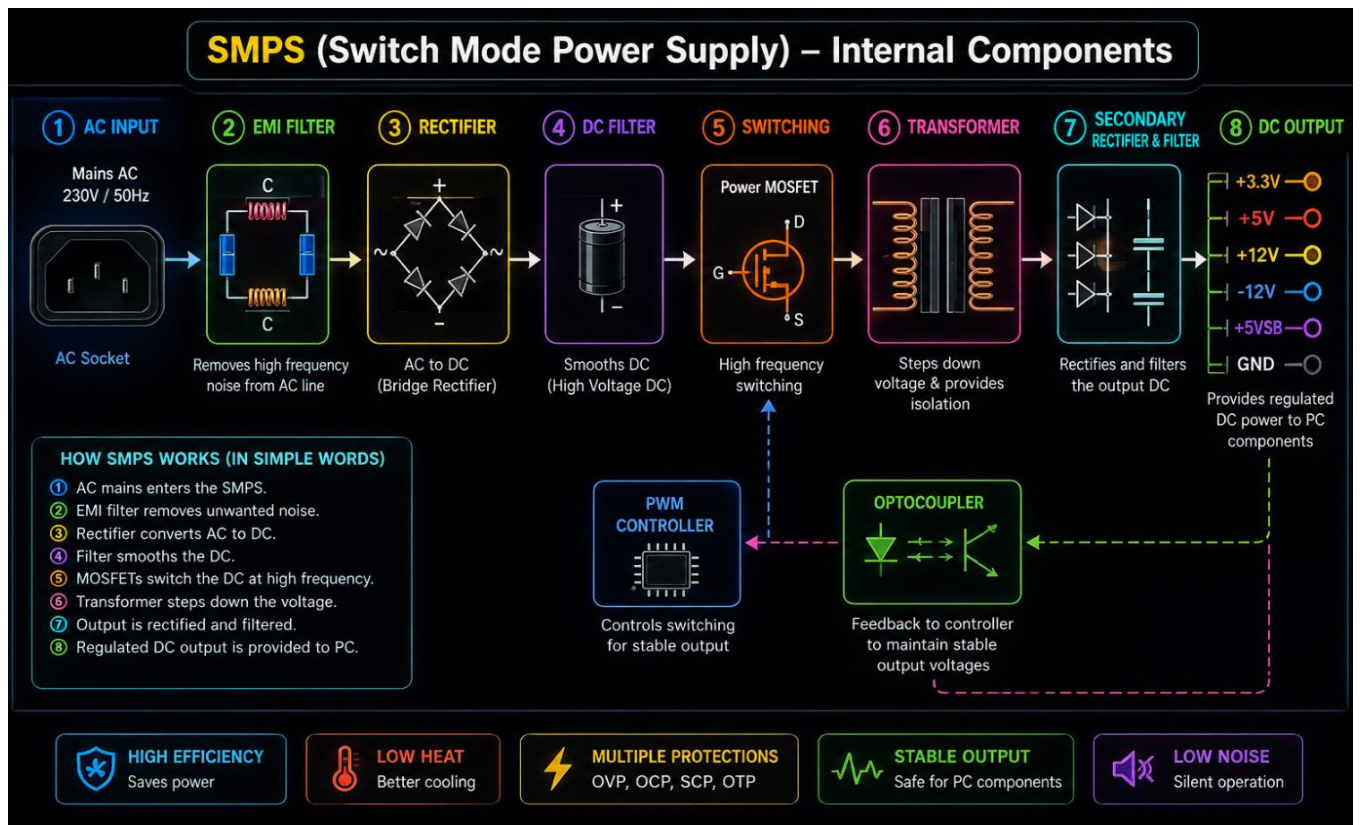
❖ Devices Using SMPS

No.	Device	Why it uses SMPS
1.	Desktop Computer	SMPS gives the computer the correct voltage and power.
2.	LED TV	SMPS provides stable power to run the TV safely.
3.	Wi-Fi Router	SMPS converts electricity into the low voltage needed by the router.

❖ Devices Using UPS

No.	Device	Why it uses UPS
1.	Computer	UPS keeps the computer running for some time when electricity goes off.
2.	CCTV Camera System	UPS keeps the cameras working during a power cut.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.



❖ SMPS – 8 Simple Steps

1. AC Input

→ Electricity enters the SMPS from the wall socket.

2. EMI Filter

→ Removes unwanted electrical noise.

3. Rectifier

→ Converts AC power into DC power.

4. DC Filter

→ Smooths the DC power.

5. Switching

→ MOSFET switches the power at high speed.

6. Transformer

→ Changes the voltage to the required level and provides isolation.

7. Secondary Rectifier & Filter

→ Converts and smooths the power again.

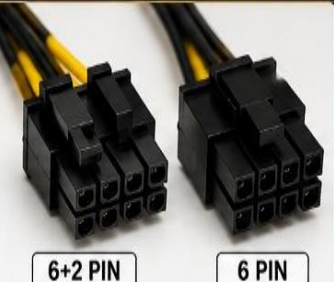
8. DC Output

→ Gives stable +3.3V, +5V and +12V power to the PC components.

❖ Easy flow:

AC → Filter → Rectifier → DC Filter → Switching → Transformer → Rectifier & Filter → DC Output.

3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

PC POWER CONNECTORS			
1. ATX 24-PIN (Motherboard Connector)	2. SATA POWER (15-Pin Connector)	3. MOLEX CONNECTOR (4-Pin Connector)	4. PCIe 6/8-PIN (GPU Power Connector)
 	 	 	 
Name: ATX 24-Pin Motherboard Connector	Name: SATA Power (15-Pin Connector)	Name: Molex Connector (4-Pin Connector)	Name: PCIe 6/8-Pin (GPU Power Connector)
Devices it Powers: Powers the motherboard. Provides power to most components on the board.	Devices it Powers: Powers SATA devices such as HDDs, SSDs, and DVD/Blu-ray drives.	Devices it Powers: Powers older devices such as HDDs, optical drives, case fans, and some accessories.	Devices it Powers: Powers graphics cards (GPU) that require additional power.
 Safety Tip: Turn off the PC and PSU before connecting or removing it. Make sure it is fully inserted for a secure connection.	 Safety Tip: Do not force it. Align the connector properly before plugging in.	 Safety Tip: Hold the connector while plugging/unplugging. Do not pull the wires.	 Safety Tip: Use the correct PCIe cable (not CPU/EPS cable). Ensure it is fully inserted.
 General Safety: Always switch off the PC and unplug the power cord before connecting or disconnecting any power connectors.			

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

SMPS Cable Connection Steps

1. **Turn OFF the PC**

→ Switch off the SMPS and unplug the power cable.

2. **Connect Motherboard**

→ Connect the **24-pin ATX cable** from SMPS to the motherboard's **24-pin ATX socket**.

3. **Connect CPU**

→ Connect the **4/8-pin CPU (EPS) cable** from SMPS to the motherboard's **CPU power socket** near the CPU.

4. **Connect Graphics Card**

→ Connect the **PCIe 6-pin / 8-pin / 6+2-pin cable** from SMPS to the GPU.

5. **Connect HDD/SSD**

→ Connect the **SATA power cable** from SMPS to the HDD or SATA SSD.

6. **Connect Other Devices**

→ If required, connect **SATA power** to DVD drives or **Molex** to older fans/accessories.

7. **Check Connections**

→ Make sure every connector is fully inserted and properly locked.

8. **Power ON**

→ Connect the main power cable, switch on the SMPS, and start the PC.